DATABASE MANAGEMENT SYSTEMS: NORMALIZATION & ACID

Chapter 1: Relational Database Normalization
Database normalization is the process of organizing relational tables to reduce data redundancy and improve data integrity.

1. First Normal Form (1NF):
- Each column must contain atomic (indivisible) values.
- No repeating groups or arrays within a record.
- Every table must possess a unique primary key.

2. Second Normal Form (2NF):
- Must satisfy all 1NF rules.
- Eliminates Partial Dependency: All non-key attributes must depend fully on the primary key, not merely a subset of a composite primary key.

3. Third Normal Form (3NF):
- Must satisfy all 2NF rules.
- Eliminates Transitive Dependency: Non-key attributes must depend directly on the primary key, and not on other non-key attributes (X -> Y and Y -> Z where Z is non-key).

4. Boyce-Codd Normal Form (BCNF):
- A stricter 3.5NF variation. For every functional dependency X -> Y, X must be a super key.

Chapter 2: Transaction Concurrency & ACID Properties

ACID guarantees safe execution of database transactions:
- Atomicity: 'All or nothing'. Either all database operations in a transaction succeed and commit, or the entire transaction is rolled back to its initial state upon failure.
- Consistency: Ensures transactions transition the database from one valid state to another, respecting all schema constraints, cascades, and foreign keys.
- Isolation: Ensures concurrent execution of transactions leaves the database in the same state as if they were executed sequentially. Standard ANSI SQL isolation levels include:
  1. Read Uncommitted
  2. Read Committed
  3. Repeatable Read
  4. Serializable
- Durability: Once a transaction commits, its effects persist permanently in non-volatile storage, even across sudden power failures or system crashes (guaranteed by Write-Ahead Logging / WAL).

Chapter 3: B+ Tree Indexing

B+ Trees are self-balancing search trees that store all actual data records in leaf nodes connected by a bidirectional linked list, while internal nodes store index navigation keys.
They maintain O(log N) lookup, insertion, and deletion complexity and provide exceptional range query performance.